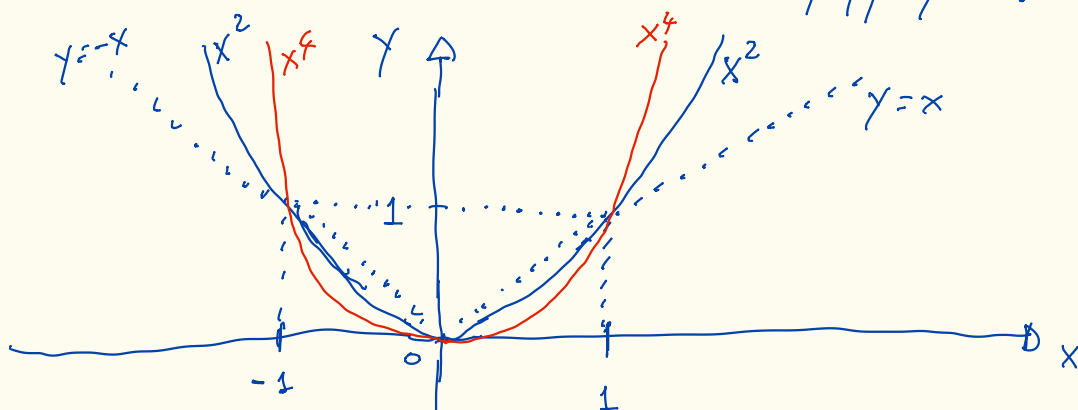


GRAFICI DI POTENZE E RADICI

$$y = x^k$$

k pari

$k = 2, 4, 6, \dots$



1) DOMINIO = $\mathbb{R}$

2) $x^2, x^4, x^6, \dots$

SONO SEMPRE ≥ 0

$$x^4 \geq x^2 \quad \text{PER } x \geq 1$$

$$x^4 \leq x^2 \quad \text{PER } 0 \leq x \leq 1$$

$$x^2 \geq x$$

PER $x \geq 1$

$$x^2 \leq x$$

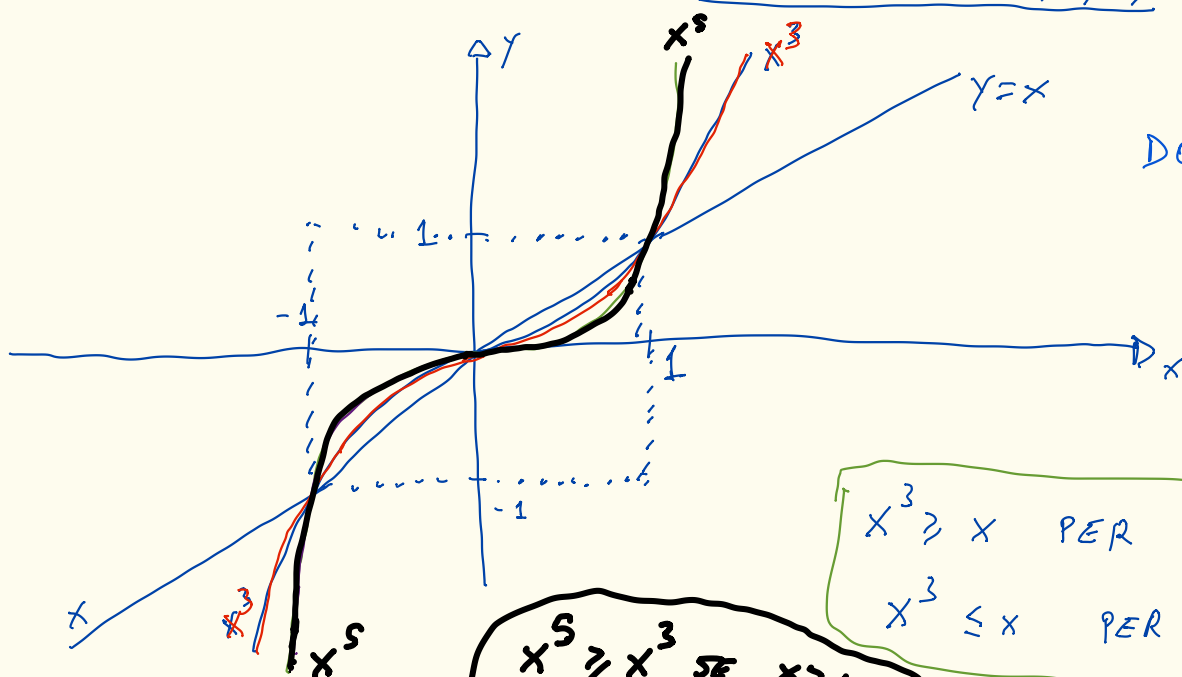
PER $0 \leq x \leq 1$

ESPOONENTI DISPARI

$$y = x^k$$

k DISPARI

$k = 1, 3, 5, \dots$



DOMINIO = $\mathbb{R}$

$$x^3 \geq x \quad \text{PER } x \geq 1$$

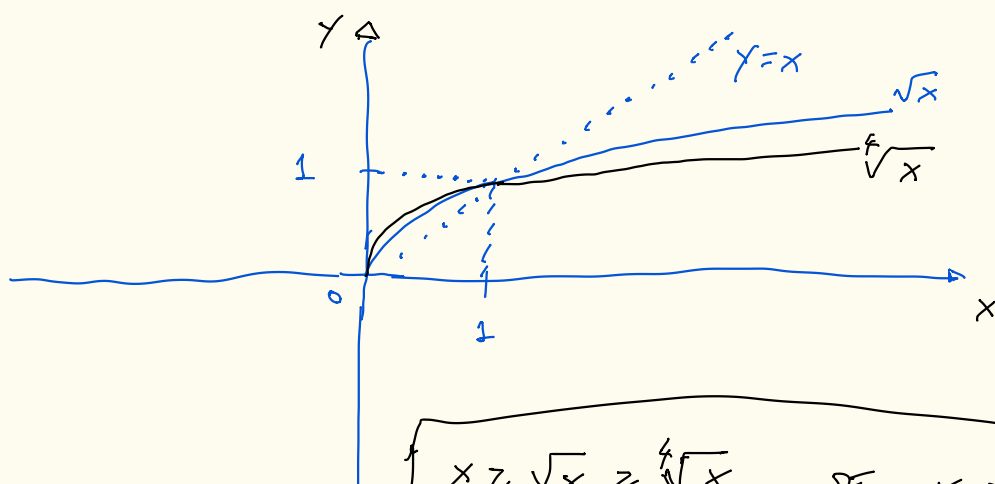
$$x^3 \leq x \quad \text{PER } 0 \leq x \leq 1$$

$$x^5 \geq x^3 \quad \text{SE } x \geq 1$$

$$x^5 \leq x^3 \quad \text{SE } 0 \leq x \leq 1$$

RADICI PARI

$$y = \sqrt{x}, \sqrt[4]{x}, \sqrt[6]{x}, \dots$$



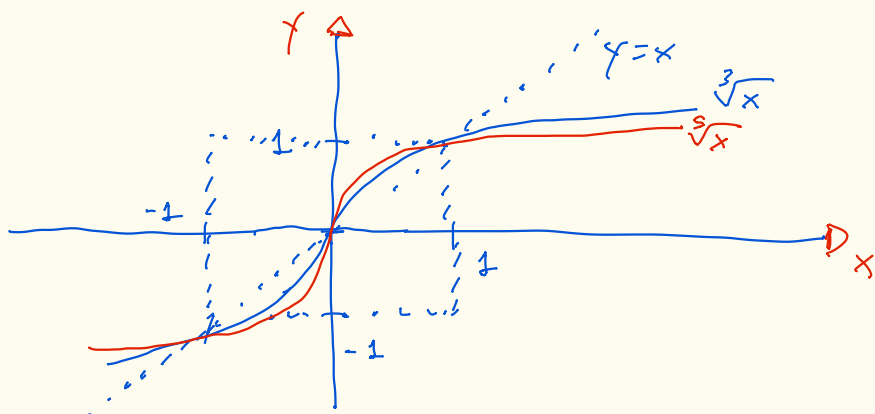
$$\text{DOMINIO} = [0, +\infty)$$

RADICI PARI SONO
SEMPRE ≥ 0

$$\begin{aligned} x &\geq \sqrt{x} \geq \sqrt[4]{x} && \text{SE } x \geq 1 \\ x &\leq \sqrt{x} \leq \sqrt[4]{x} && \text{SE } 0 \leq x \leq 1 \end{aligned}$$

RADICI DISPARI

$$y = \sqrt[3]{x}, \sqrt[5]{x}, \dots$$



$$\text{DOMINIO} = \mathbb{R}$$

ESPONENZIALI E LOGARITMI

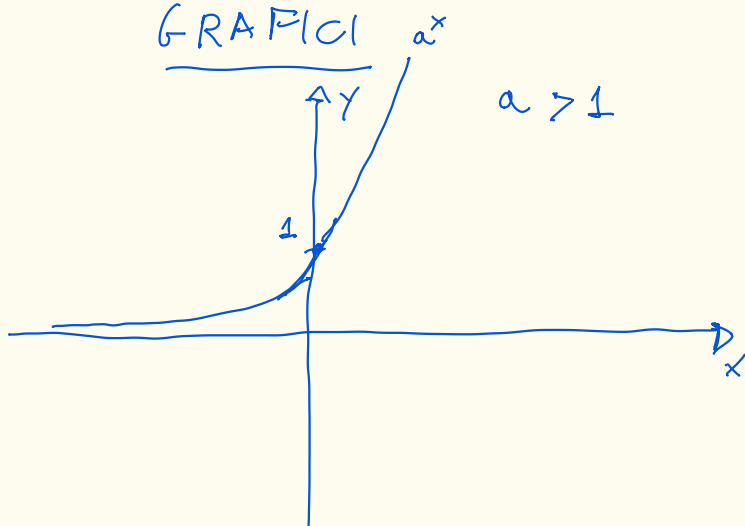
ESPONENZIALI

$$y = a^x$$

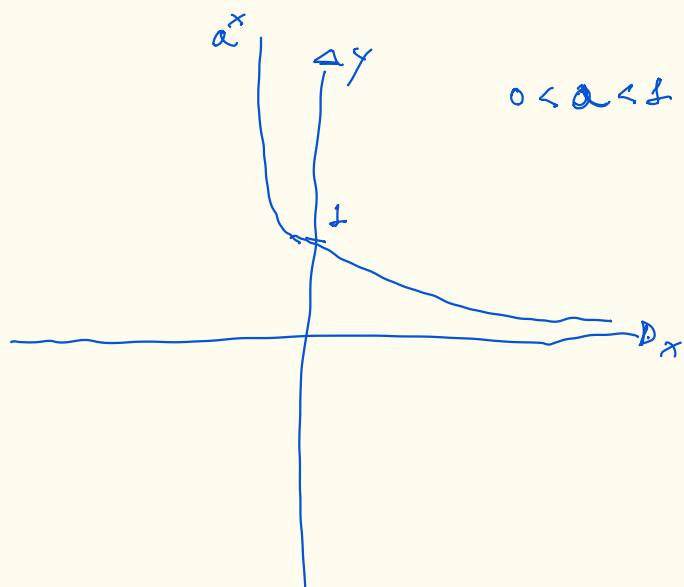
a = BASE

$$a > 0, \quad a \neq 1$$

GRAFICI



$$y = 2^x, 3^x, e^x, \dots$$



$$y = \left(\frac{1}{2}\right)^x, \left(\frac{1}{3}\right)^x, \dots$$

DOMINIO = $\mathbb{R}$ SEMPRE POSITIVE

$$a^0 = 1$$

LOGARITMI

DATA UNA BASE $a > 0$, $a \neq 1$
E $y > 0$ ESISTE UN UNICO REALE x TALE CHE

$$a^x = y$$

SI DICE CHE $x = \log_a(y)$

ESEMPIO

$$\log_2(8) = 3 \quad \text{PERCHÉ} \quad 2^3 = 8$$

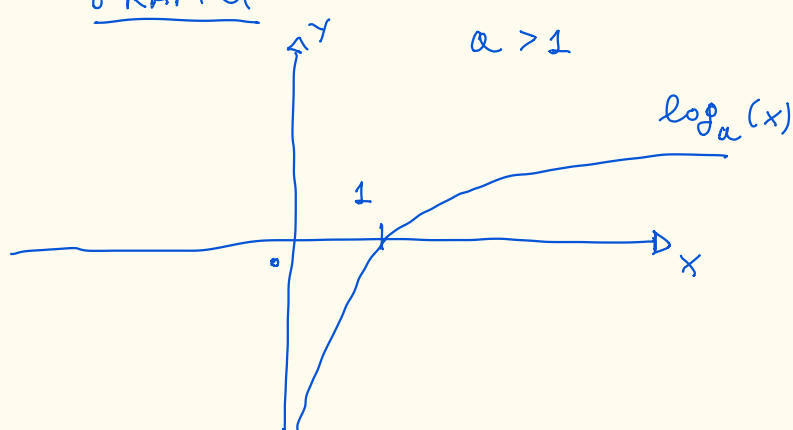
$$\log_4(64) = 3 \quad \text{PERCHÉ} \quad 4^3 = 64$$

$$\log_2\left(\frac{1}{8}\right) = -3$$

$$\log_a(1) = 0 \quad (a^0 = 1)$$

$$\log_a(a) = 1 \quad (a^1 = a)$$

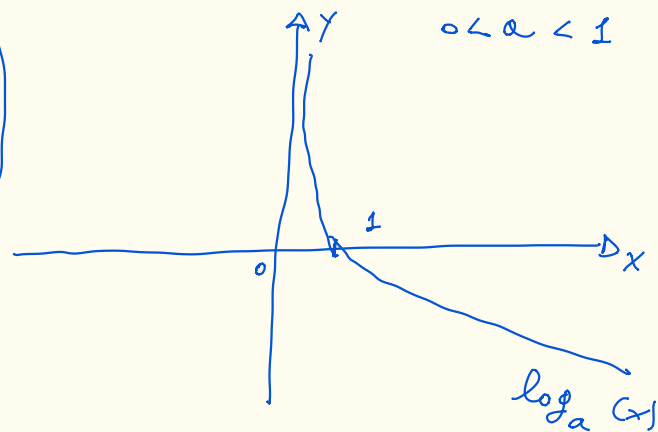
GRAFICI



$$\text{DOMINIO} = (0, +\infty)$$

$$\log_a(x) \geq 0 \quad \text{SE } x \geq 1$$

$$\log_a(x) \leq 0 \quad \text{SE } 0 < x \leq 1$$



$$\text{DOMINIO} = (0, +\infty)$$

$$\log_a(x) \geq 0 \quad \text{SE } 0 < x \leq 1$$

$$\log_a(x) \leq 0 \quad \text{SE } x \geq 1$$

e = NUMERO DI NEPERO

$$e = 2,718 \dots$$

REALE MA NON RAZIONALE

IL LOGARITMO IN BASE e È DETTO LOGARITMO NATURALE

E SI INDICA CON $\ln(x)$:

$$\text{QUINDI } \log_e(x) \doteq \ln(x)$$

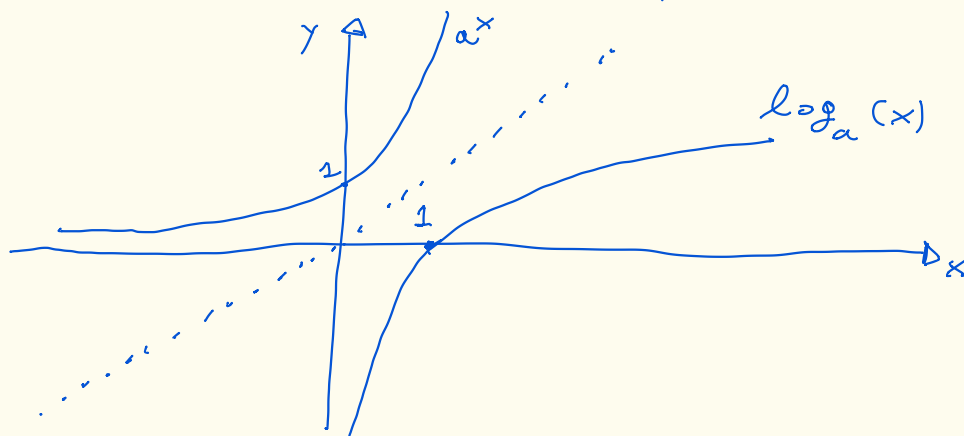
RELAZIONE DI INVERTIBILITÀ

a^x È UNA FUNZIONE CON DOMINIO $\mathbb{R}$,
IMMAGINE $(0, +\infty)$ E INIETTIVA

QUINDI ESISTE L'INVERSA CON DOMINIO $(0, +\infty)$

E IMMAGINE $\mathbb{R}$. L'INVERSA DI a^x È PROPRIO $\log_a(x)$

I GRAFICI DI FUNZIONI INVERSE SONO SIMMETRICI
RISPETTO ALLA BISETTRICE $y=x$



RELAZIONI DI INVERTIBILITÀ

UTILI
PER
RISOLVERE
EQUAZIONI



$$\log_a(x) = x \quad \text{SE } x > 0$$



$$\log_a(a^x) = x \quad \forall x \in \mathbb{R}$$

PERCHÉ SE $g(y)$ È L'INVERSA DI $f(x)$

ALLORA

$$g(f(x)) = x$$

E

$$f(g(y)) = y$$

PROPRIETÀ DEL $\log_a(x)$

$$1) \log_a(xy) = \log_a(x) + \log_a(y)$$

$$2) \log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$$

$$3) \log_a(x^\alpha) = \alpha \cdot \log_a(x) \quad \alpha \in \mathbb{R}$$

$$4) \text{CAMBIAMENTO DI BASE} \quad \log_b(x) = \frac{\log_a(x)}{\log_a(b)}$$

DIMOSTRAZIONE DI (1)

$$\log_a (xy) = z_1 \quad \text{VUOL DIRE}$$

$$a^{z_1} = xy$$

$$\log_a (x) = z_2$$

$\longleftrightarrow$

$$\boxed{a^{z_2} = x}$$

$$\log_a (y) = z_3$$

$\longleftrightarrow$

$$\boxed{a^{z_3} = y}$$

SOSTITUISCO

$$\boxed{a^{z_1}} = a^{z_2} a^{z_3} = \boxed{a^{z_2 + z_3}}$$

$$\Rightarrow z_1 = z_2 + z_3$$

$$\Rightarrow \log_a (xy) = \log_a (x) + \log_a (y)$$

EQUAZIONI

$$2^x = 64$$

APPLICO $\log_2$ AD ENTRAMBI I MEMBRI

$$\log_2 (2^x) = \log_2 (64)$$

$\nearrow$ //

//

CONDIZ
DI INVERTI
BILITA'

x

=

6

$$2^6 = 64$$

$$4^x = 4^{x-2} + 27$$

$$4^x - 4^{x-2} = 27$$

$$4^x - 4^x \cdot 4^{-2} = 27$$

$$4^x - \frac{1}{16} 4^x = 27$$

$$4^x \left(1 - \frac{1}{16} \right) = 27$$

$$4^x \left(\frac{15}{16} \right) = 27$$

DIVIDO PER $\frac{15}{16}$

$$4^x = \frac{16}{15} \cdot 27 = \frac{16}{5} \cdot 9$$

APPLICO $\log_4$

$$x = \log_4(4^x) = \log_4 \left(\frac{16 \cdot 9}{5} \right)$$

$$5^x \cdot 2^{2x} = 10$$

$$2^{2x} = (2^2)^x = 4^x$$

$$5^x \cdot 4^x = 10$$

$$20^x = 10$$

$$x = \log_{20}(20^x) = \log_{20}(10)$$

ESERCIZIO PER CASA

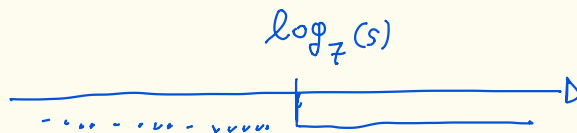
$$21^{x-1} = 15^x$$

DISEQUAZIONI

$$7^x > 5$$

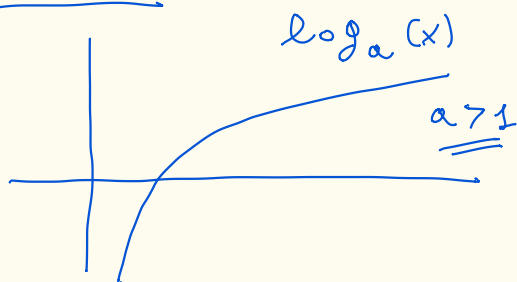
APPLICO $\log_7$

$$x = \log_7 (7^x) > \log_7 (5)$$



$$\text{SOL} = (\log_7 (5), +\infty)$$

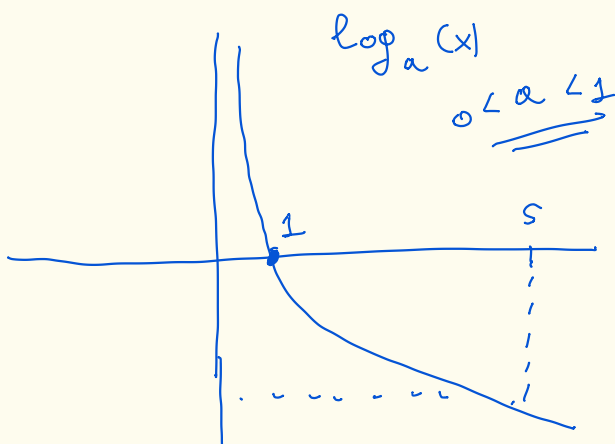
ATTENZIONE



È MONOTONA CRESCENTE:

SE $x_1 \leq x_2$ ALLORA

$$\log_a(x_1) \leq \log_a(x_2)$$



È MONOTONO DECRESCENTE:

SE $x_1 \leq x_2$ ALLORA

$$\log_a(x_1) \geq \log_a(x_2)$$

! IL VERSO CAMBIA!

ESEMPIO

$$\left(\frac{1}{2}\right)^{x^2} > 5$$

APPLICO $\log_{1/2}$

$$\log_{1/2} \left(\frac{1}{2}\right)^{x^2} \leq \log_{1/2} (5)$$

$a = \frac{1}{2} < 1$
↑
CAMBIO IL VERSO

$$x^2 \leq \log_{1/2}(s)$$

$\uparrow$ $\uparrow$
 È SEMPRE ≥ 0 È NEGATIVO

$$\text{sol} = \emptyset$$

$$e^{2x} - 4e^x + 1 < 0$$

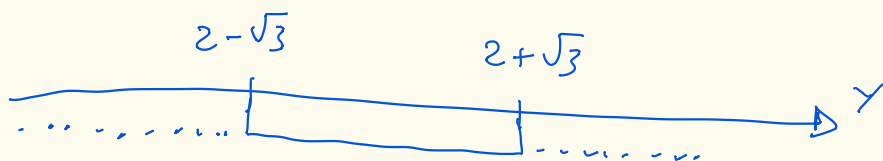
$e = \text{Neper} = 2,718\dots$

SOSTITUZIONE: $y = e^x$

$$e^{2x} = (e^x)^2 = y^2$$

$$y^2 - 4y + 1 < 0$$

SOLUZIONI $y_{1,2} = 2 \pm \sqrt{4 - 1} = 2 \pm \sqrt{3}$

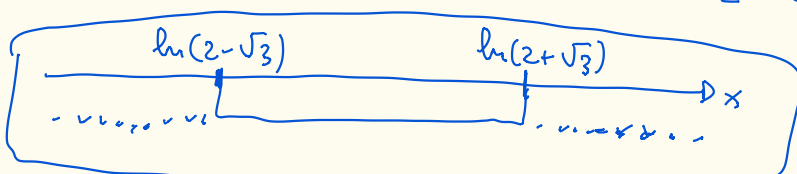


SE $y = 2 - \sqrt{3}$ ALLORA $e^x = 2 - \sqrt{3}$

APPLICO $\ln$
ENTRambi i MEMBRI

$$x = \ln(2 - \sqrt{3})$$

SE $y = 2 + \sqrt{3} = e^x \rightarrow x = \ln(2 + \sqrt{3})$



EQUAZIONI CON log

$$\ln(x^2 - 4) = \ln(x + 2)$$

CAMPO DI ESISTENZA:

$$\begin{cases} x^2 - 4 > 0 \\ x + 2 > 0 \end{cases}$$

APPLICO e^x ENTRAMBI I MEMBRI

$$\cancel{e}^{\ln(x^2 - 4)} = e^{\ln(x + 2)}$$

CONDIZIONE DI INVERT.

$$x^2 - 4 = x + 2$$

$$x^2 - x - 6 = 0$$

$$x_{1,2} = \frac{1 \pm \sqrt{1 + 24}}{2}$$

$$= \begin{cases} 3 \\ -2 \end{cases}$$

SODDISFA CAMPO DI ESISTENZA
NON SODDISFA C.E. DA SCARTARE.

$$\text{SOL} = \{3\}$$

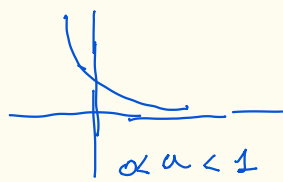
DISEQUAZIONI

$$\log_{1/2}(3x) \geq 3$$

C.E.	$3x > 0$
	$x > 0$

APPLICO

$$\left(\frac{1}{2}\right)^x$$



MONO TONIA
DECRESCENTE

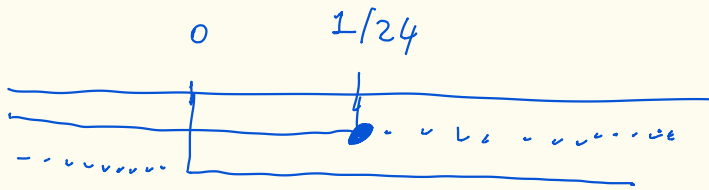
CAMBIA
VERSO

$$\left(\frac{1}{2}\right)^{\log_{1/2}(3x)}$$

$$\leq \left(\frac{1}{2}\right)^3$$

$$3x \leq \frac{1}{8}$$

$$x \leq \frac{1}{24}$$



CAMPO DI ESISTENZA

CONSIDERO I TRATTI ENTRAMBI CONTINUI

$$sol = [0, 1/24]$$

$$\log_3(x+1) > \log_9(2x+3)$$

MI RIDUCO ALLA STESSA BASE

$$\log_9(2x+3) = \frac{\log_3(2x+3)}{\log_3(9)}$$

$$= \frac{\log_3(2x+3)}{2}$$

$$\hookrightarrow \log_3 (x+1) > \frac{1}{2} \log_3 (2x+3)$$

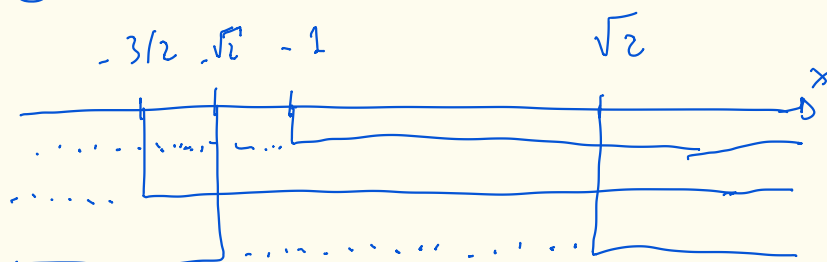
$$\rightarrow \log_3 (x+1) > \log_3 (2x+3)$$

$$\begin{cases} x+1 > 0 & \text{C.E.} \\ 2x+3 > 0 & \text{C.E.} \\ \frac{2 \log_3 (x+1)}{3} > \log_3 (2x+3) \end{cases}$$

$$\begin{cases} x > -1 \\ x > -3/2 \\ \log_3 (x+1)^2 > \log_3 (2x+3) \end{cases}$$

$$\begin{cases} x > -1 \\ x > -\frac{3}{2} \\ (x+1)^2 > 2x+3 \end{cases}$$

$$\begin{cases} x > -1 \\ x > -3/2 \\ x^2 - 2 > 0 \end{cases}$$



$$\text{SOLUTION} = (\sqrt{2}, +\infty)$$